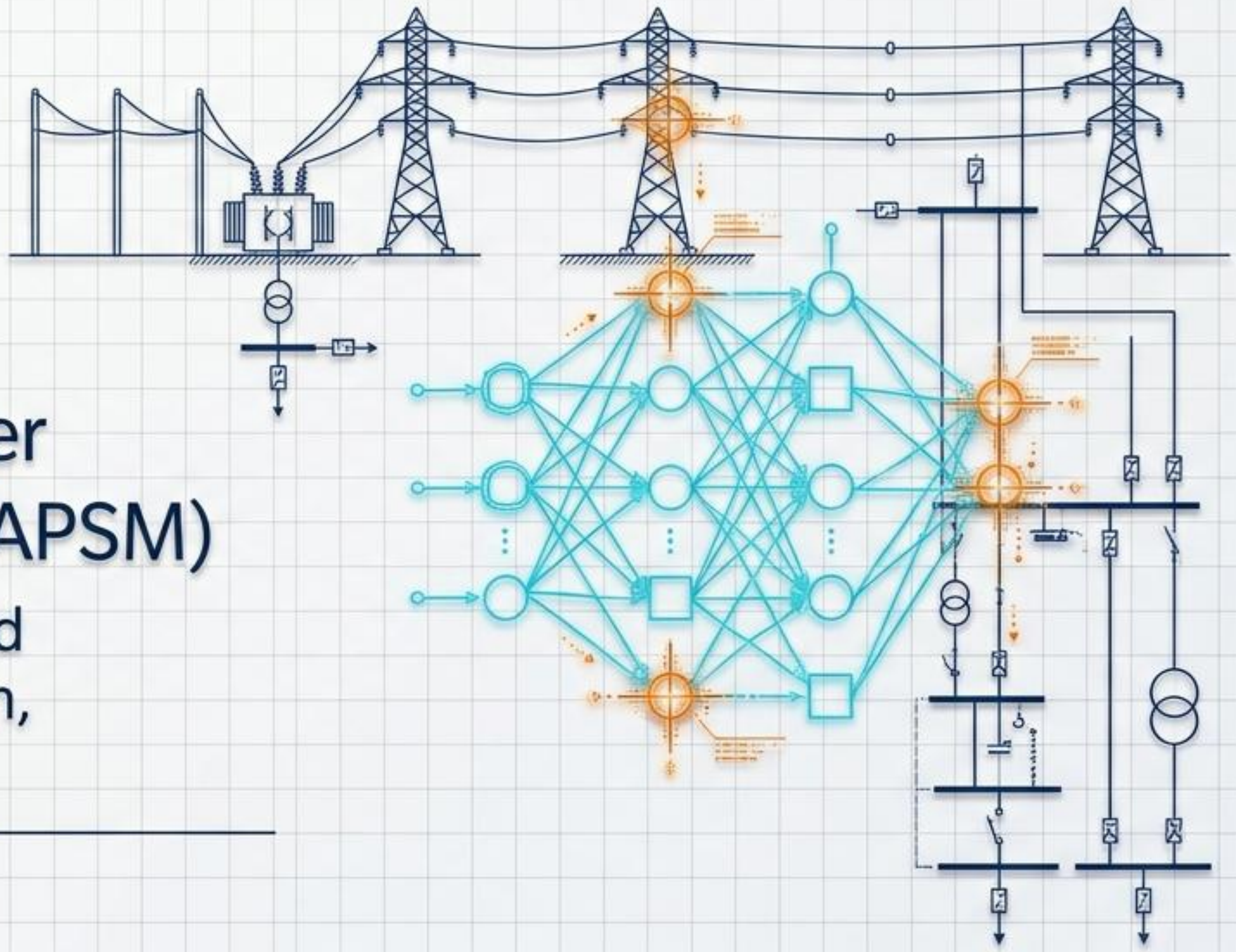
Cognitive Adaptive Power System Management (CAPSM)

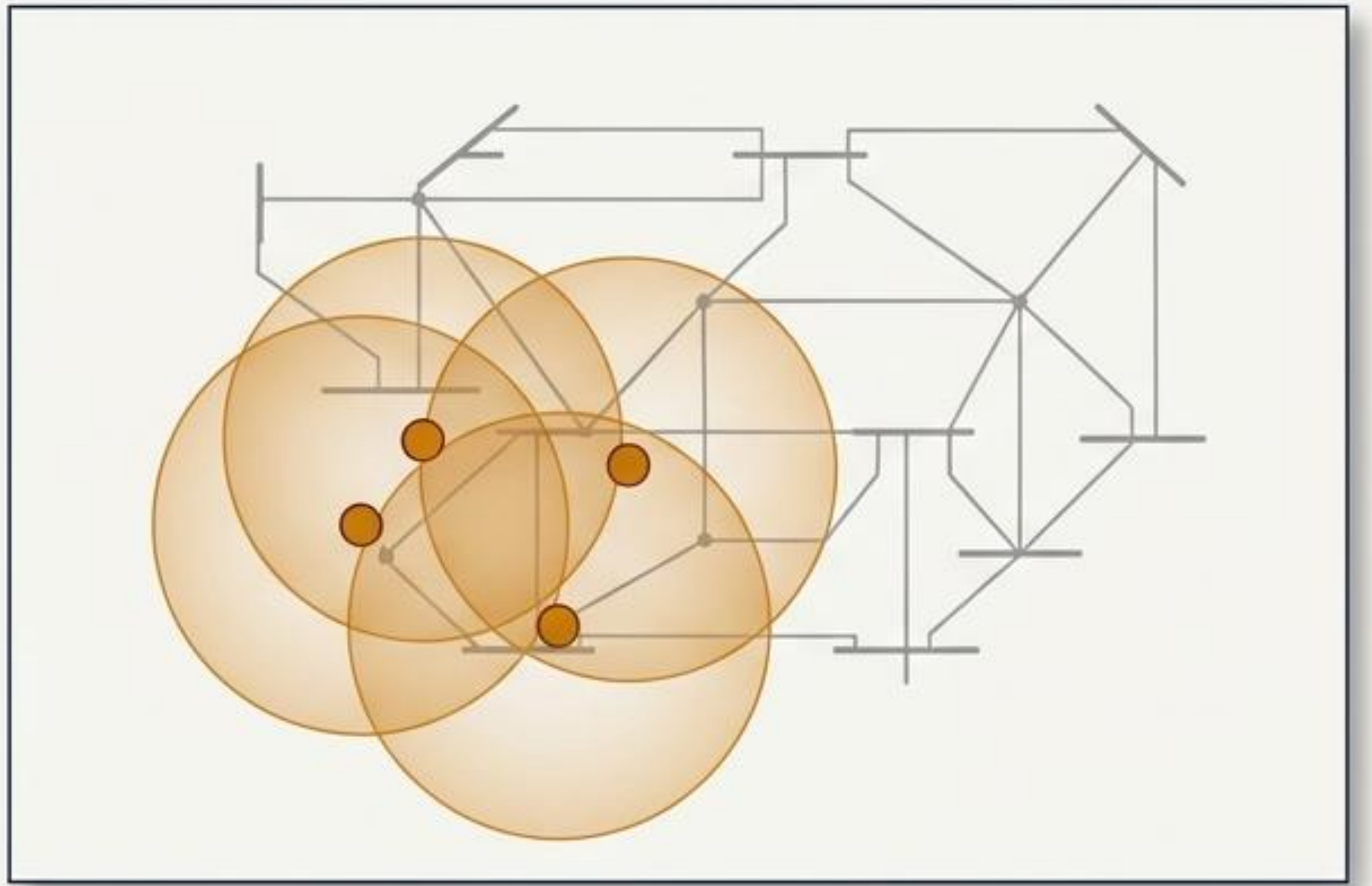
AI-Driven Optimal Placement and Physics-Informed Fault Detection, Validated on Hardware



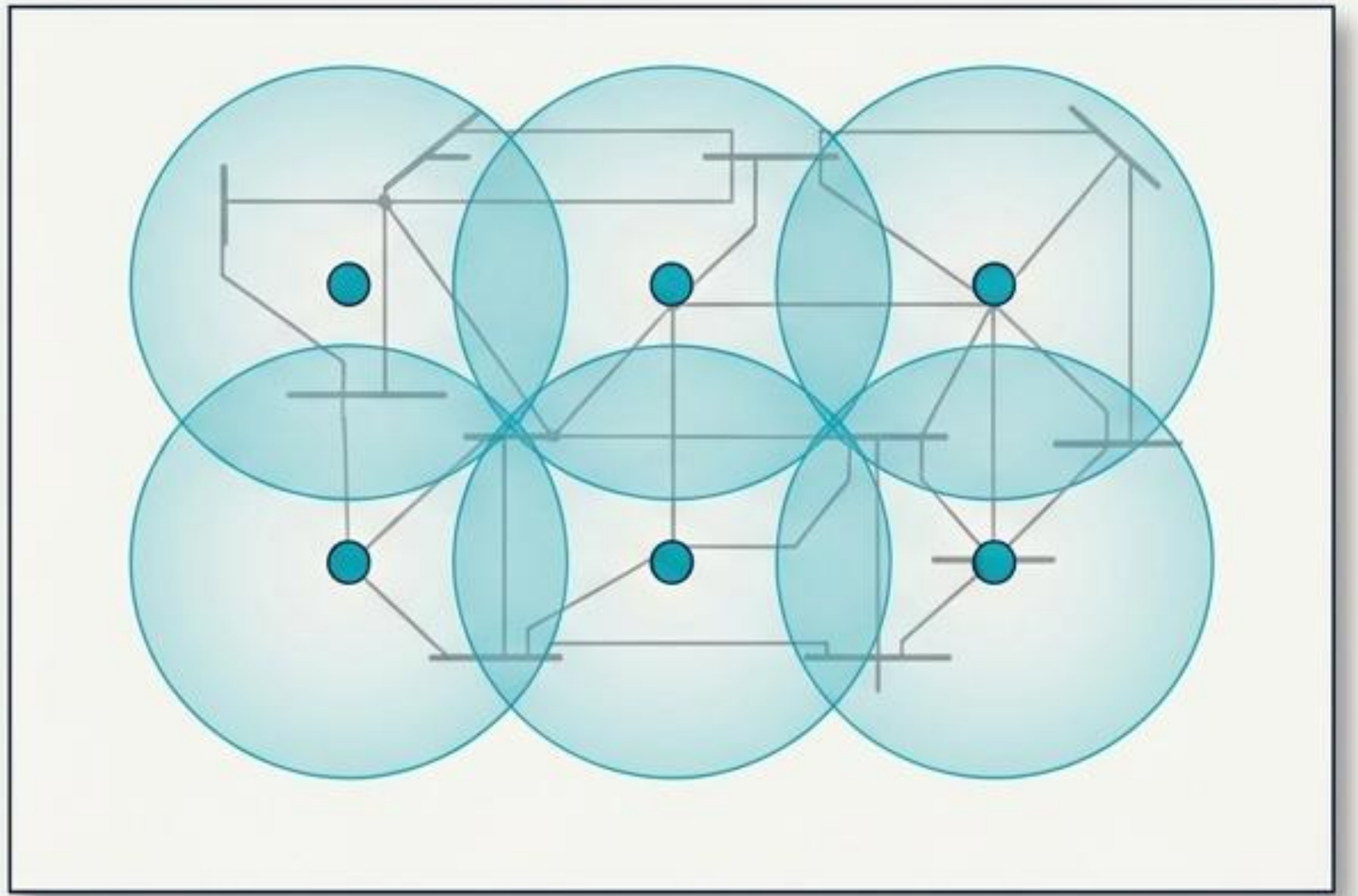
Mahmoud Kiasari
Dalhousie University

AI-Based Optimal Placement Overview

Random Placement



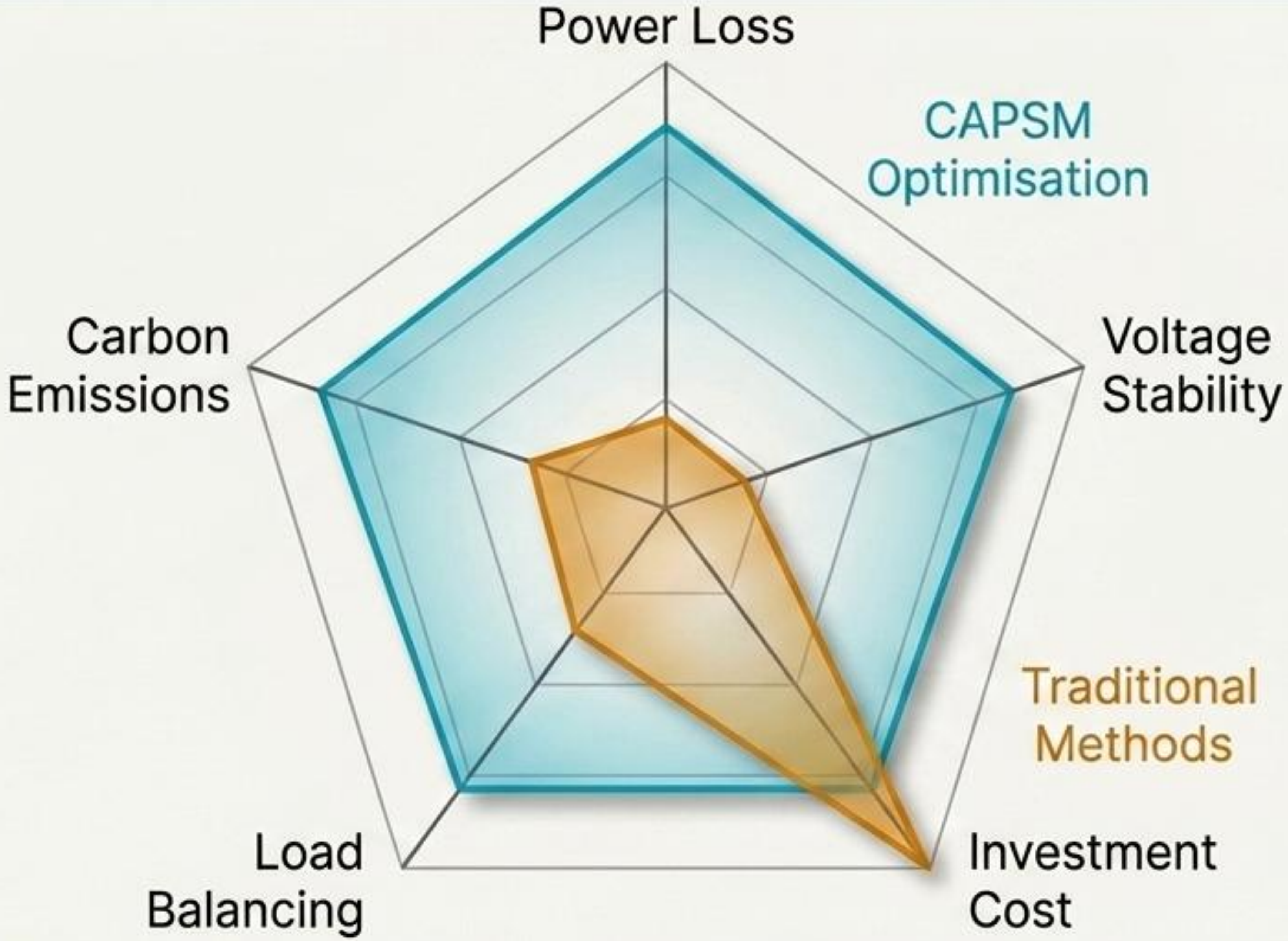
CAPSM AI-Optimised Placement



Control Authority

AI-optimised placement transforms **physical grid actuators** (FACTS devices) from isolated components into a unified, grid-wide control fabric, maximizing system influence with the same number of devices.

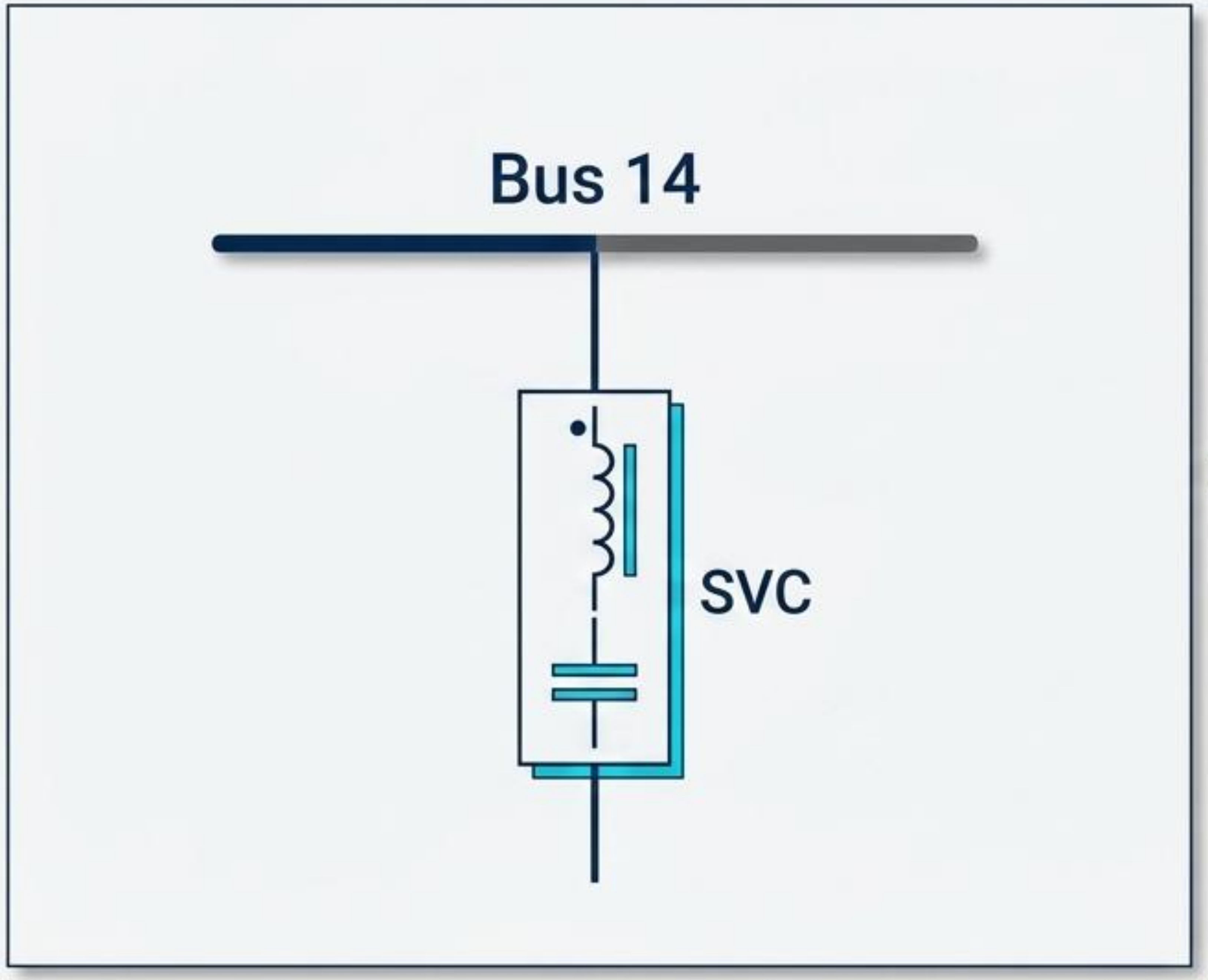
Problem Formulation: Multi-Objective Optimization

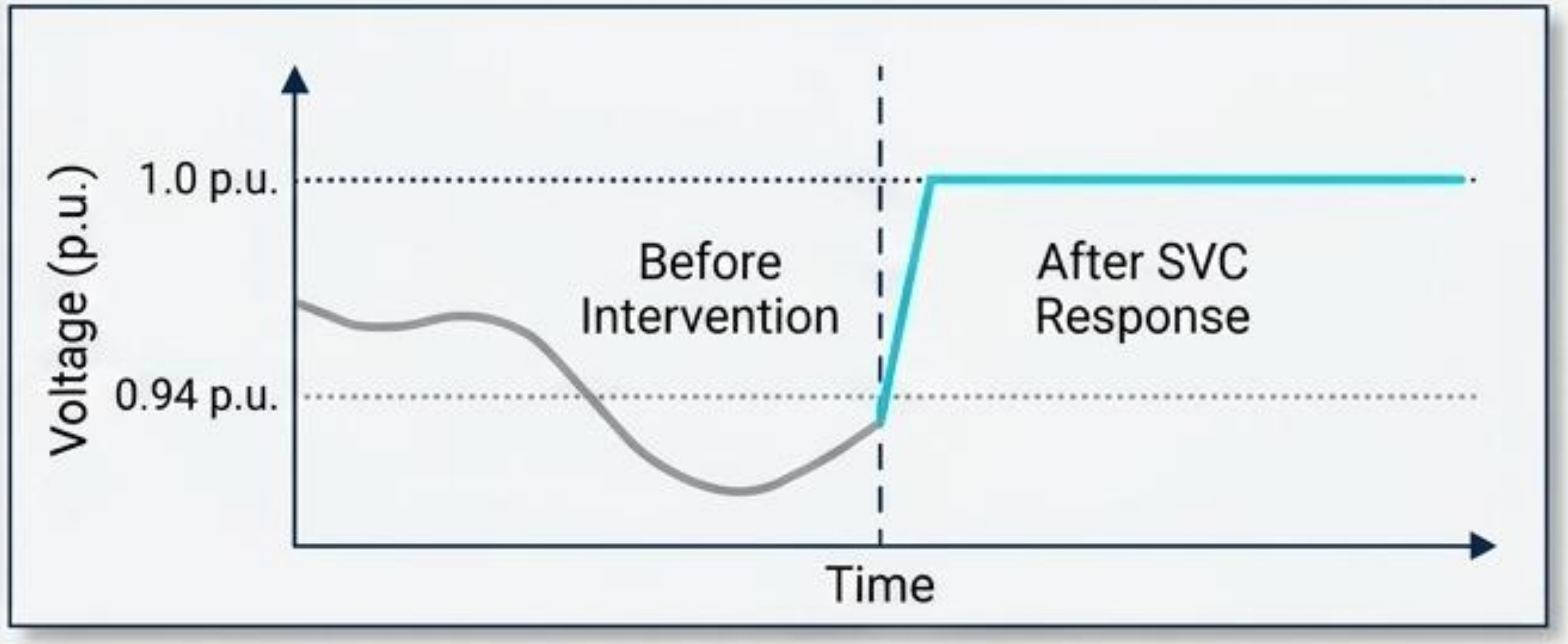


$$\min F(x,y,z) = [f_1, \ldots, f_m]$$

CAPSM resolves the tension between competing physical grid requirements by balancing five distinct operational objectives simultaneously.

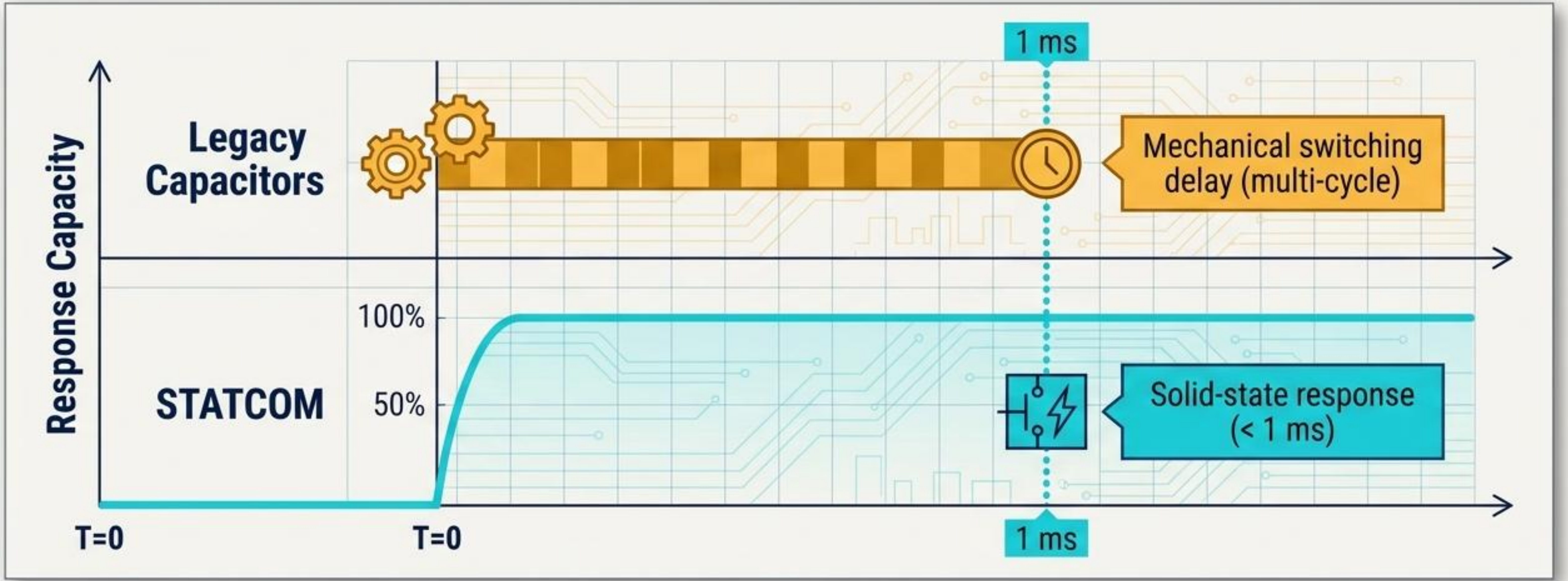
Actuator 1: SVC at Bus 14





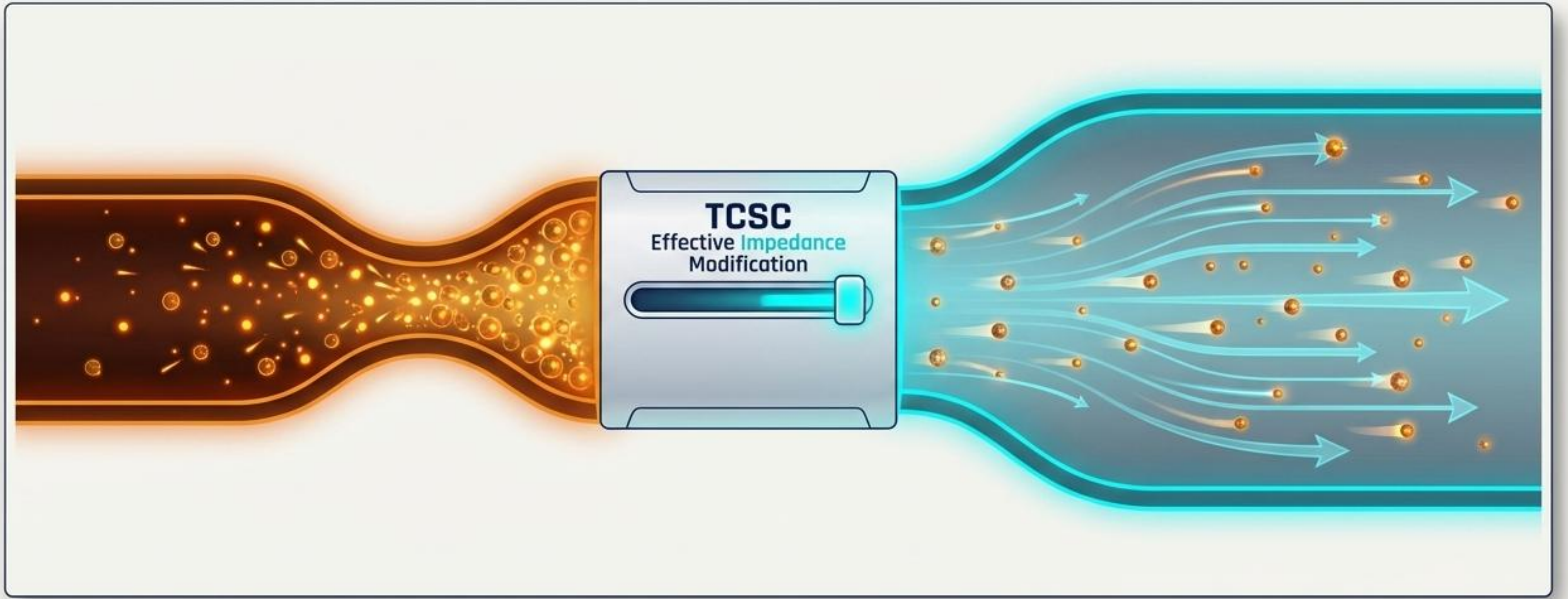
Function: Shunt Voltage Support	**Mechanism:** Dynamic Reactive Power Injection	**Target:** Bus 14

Actuator 2: STATCOM at Bus 39



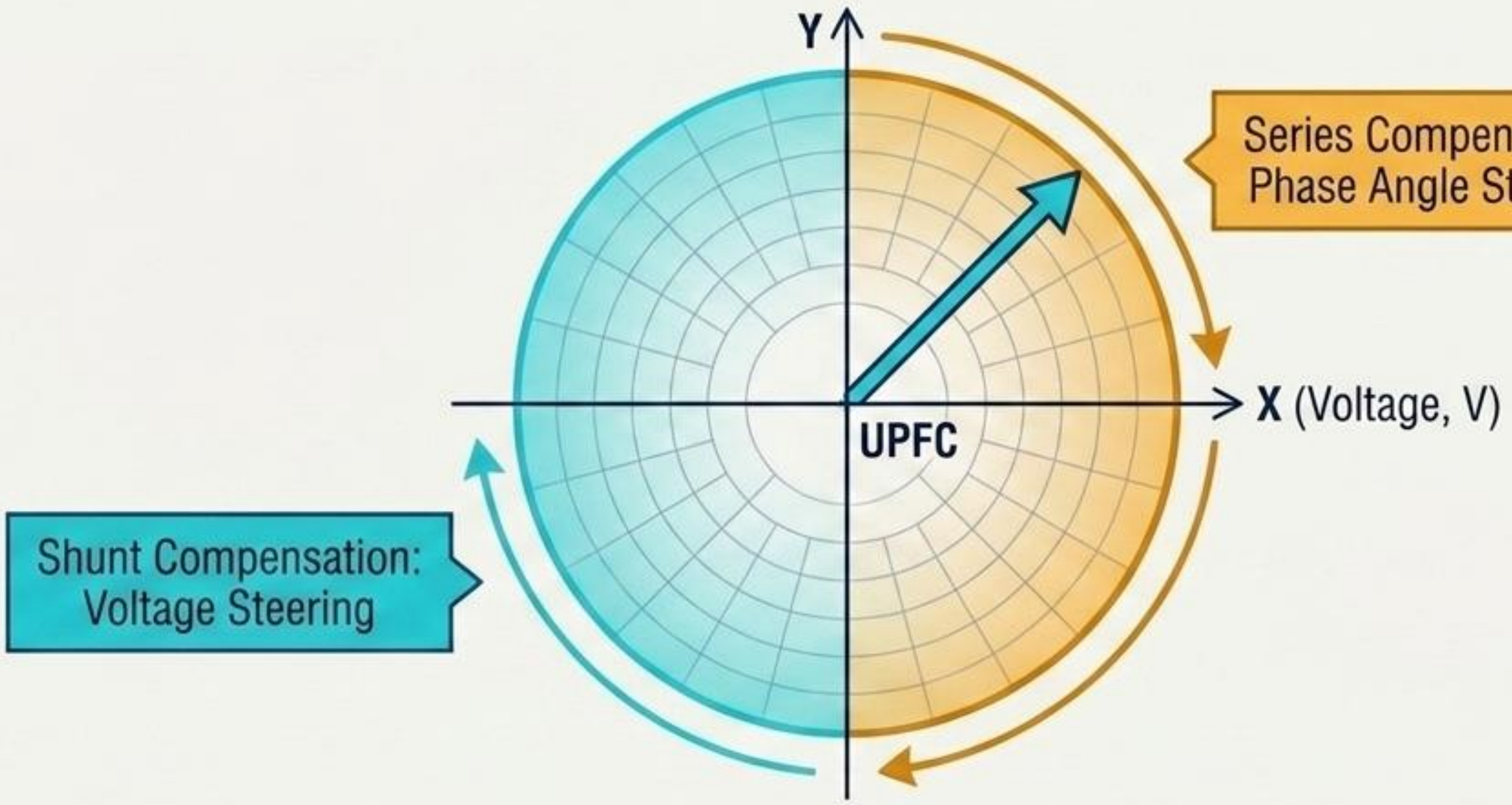
Function: Fast Reactive Power Compensation | **Advantage:** Sub-cycle solid-state response vs mechanical switching delay | **Target:** Bus 39

Actuator 3: TCSC on Line 16-17



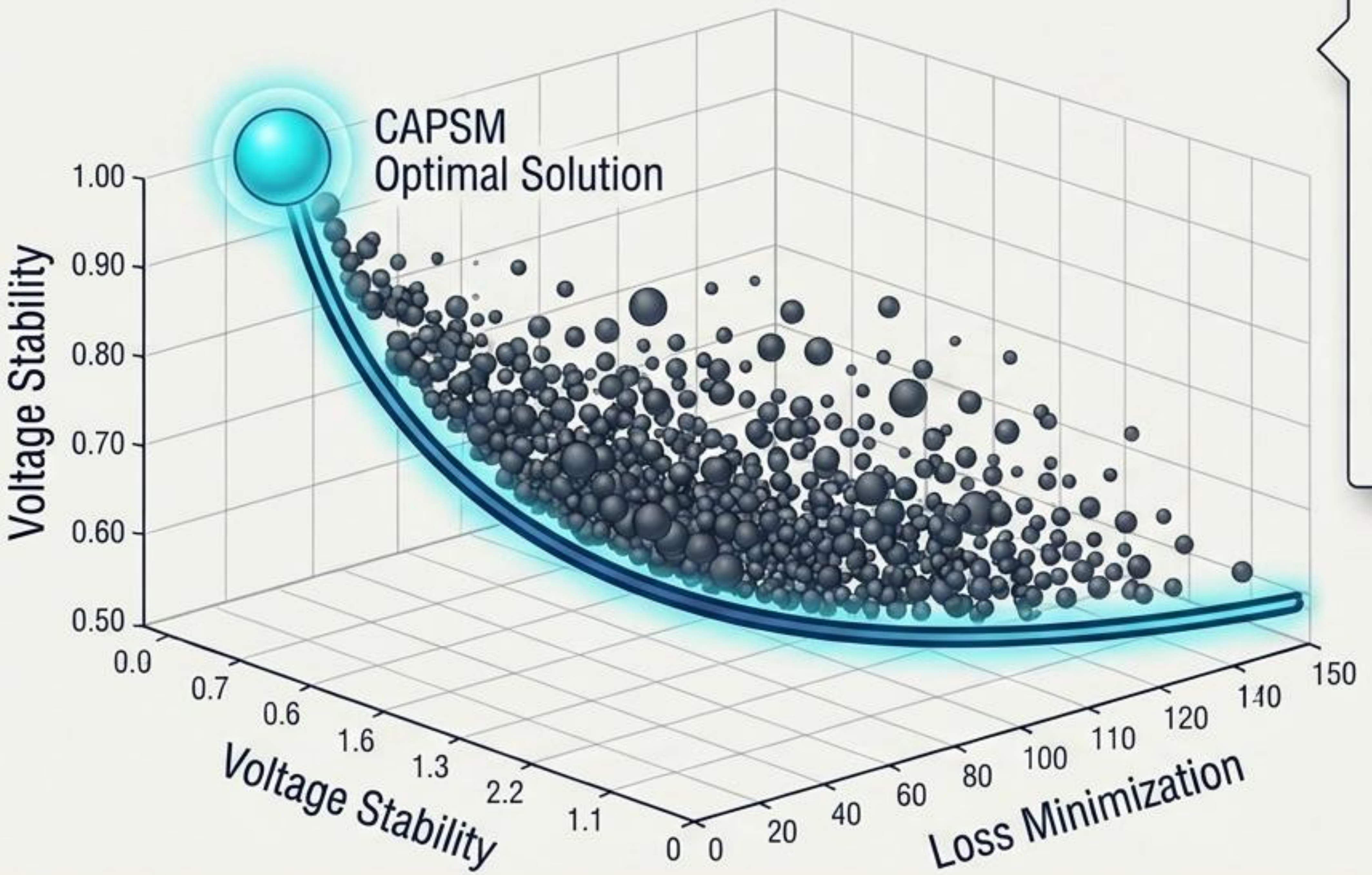
Function: Transmission Capacity Enhancement	**Mechanism:** Series effective impedance modification	**Target:** Line 16-17

Actuator 4: UPFC at Bus 26



Function: Unified Power Flow Control	**Mechanism:** Simultaneous Series + Shunt compensation	**Target:** Bus 26

The Multi-Objective Trade-off

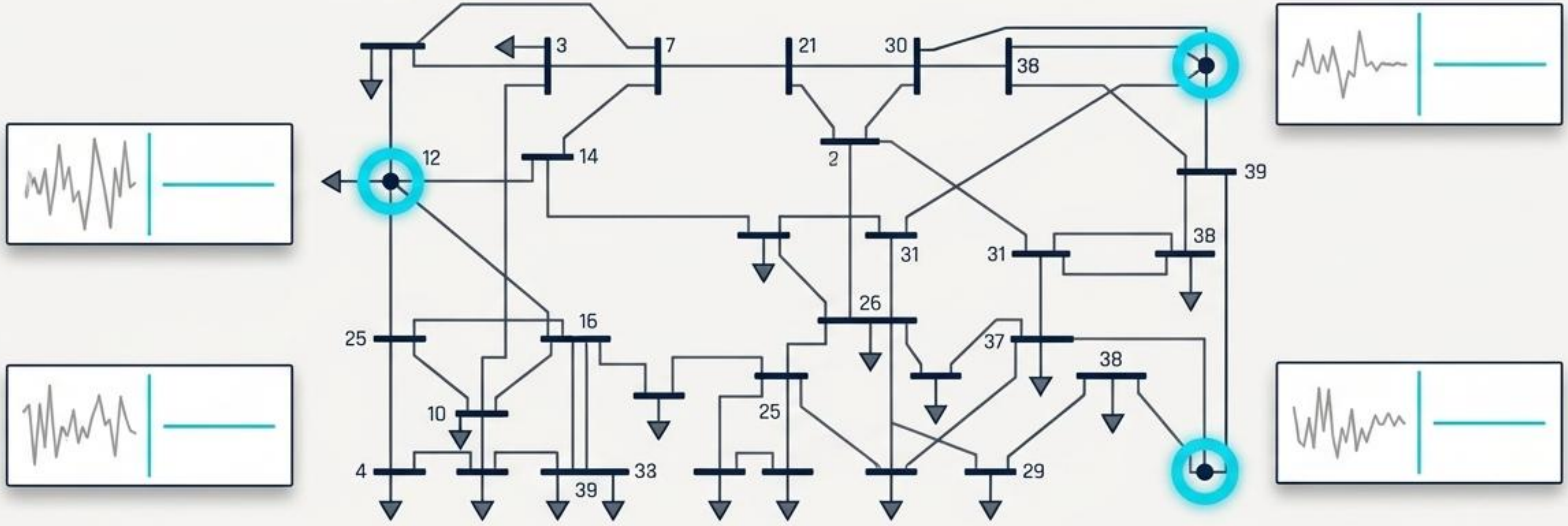


Quantum-Inspired Logic

Quantum-Inspired Reinforcement Learning (QIRL) converged 58% to 59% faster than classical DQN.

Tunneling algorithms escaped local optima to locate the absolute best configuration for the physical grid assets.

Placement Results on IEEE 39-Bus System

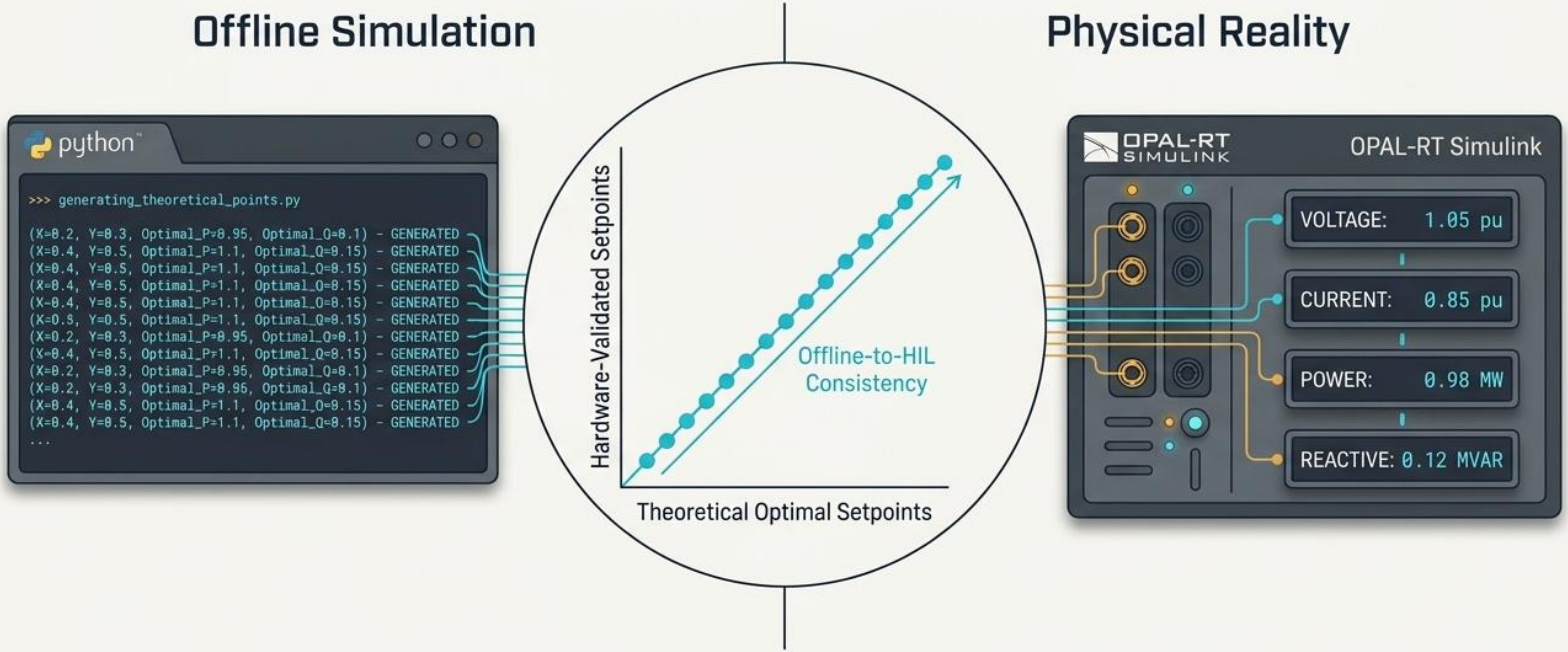


+ 37.2%
Improvement in voltage stability

- 24.3%
Decrease in system losses

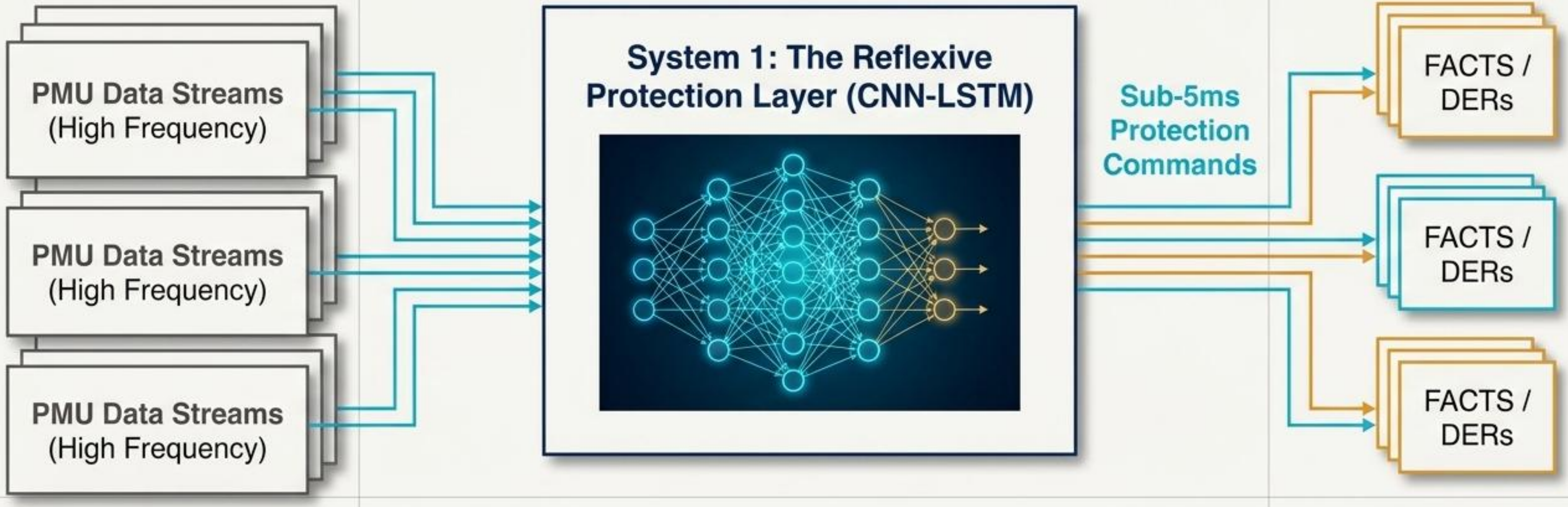
+ 28.5%
Increase in loading margin

HIL-Validated Placement Methodology



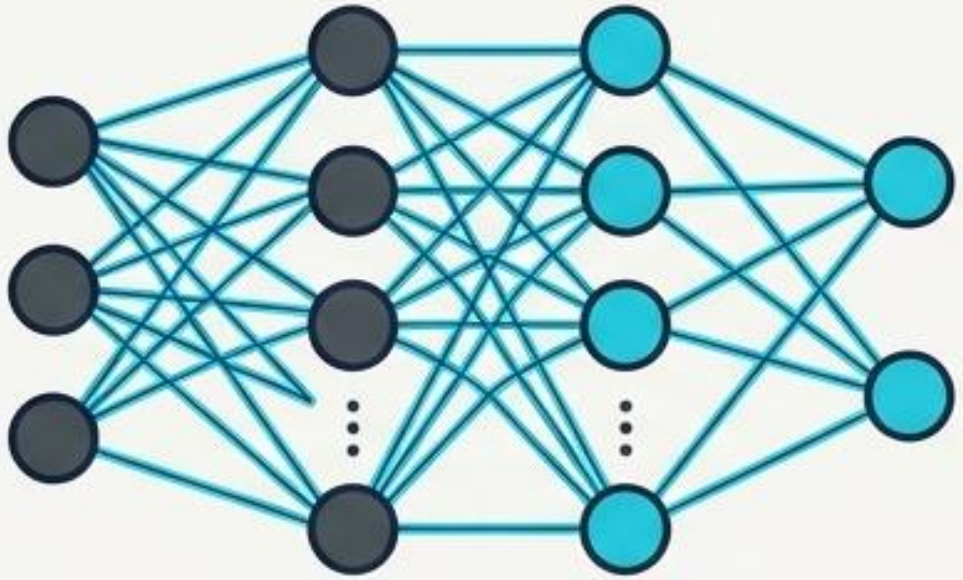
Hardware-in-the-Loop (HIL) validation catches modelling errors invisible in offline simulation, successfully bridging the gap between theoretical AI models and practical grid deployment.

Fault Detection Overview (CAPSM System 1)



This system maps human cognitive theory to grid control, acting as an involuntary, hyper-fast flinch mechanism to protect physical assets before severe damage occurs.

PINN Architecture & Physics Integration



Physics Constraint Bounding Box

$0.94 < V < 1.06$ p.u.

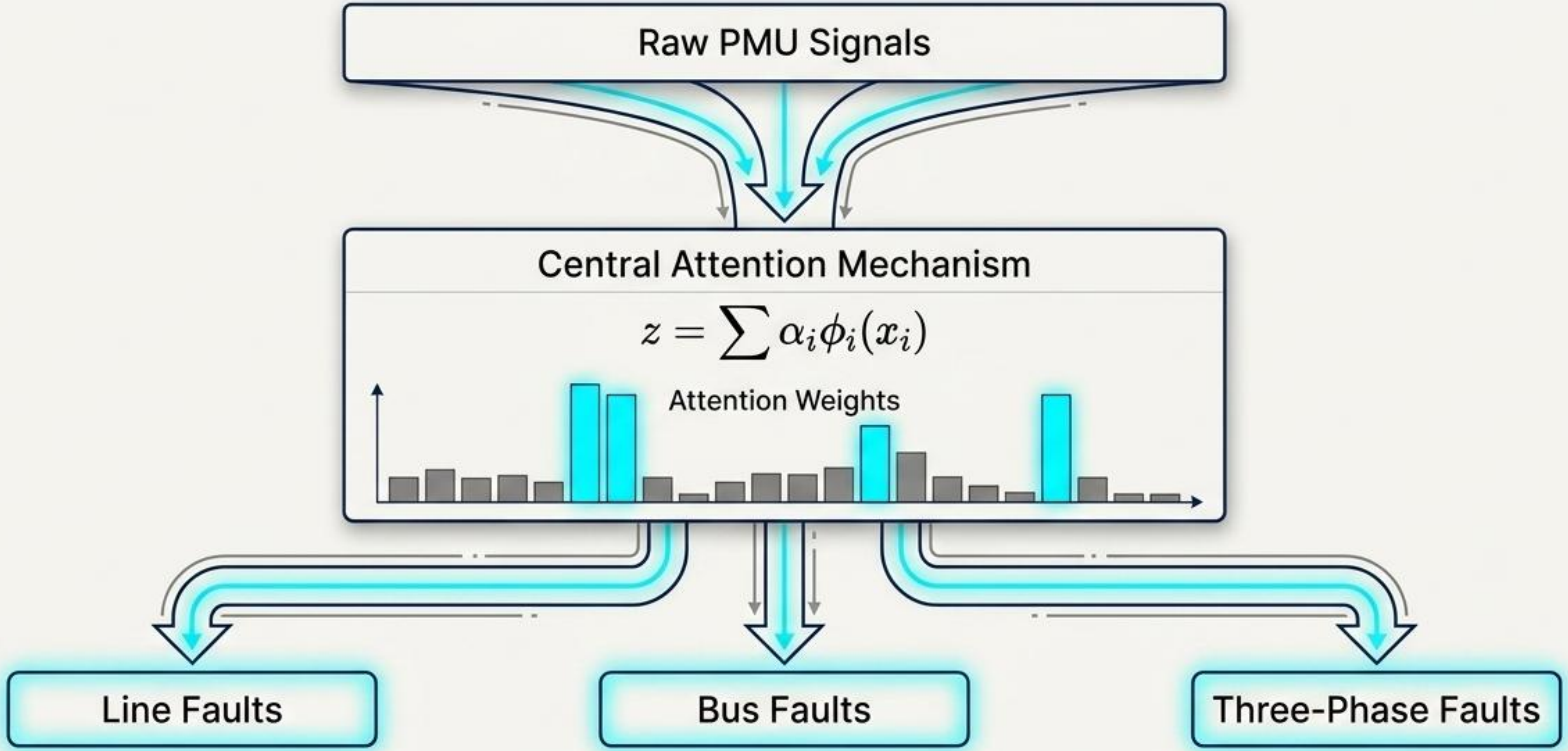
$$L_{total} = L_{data} + \lambda L_{physics}$$

Impossible outputs are mathematically penalized.

99.97%
Constraint Compliance

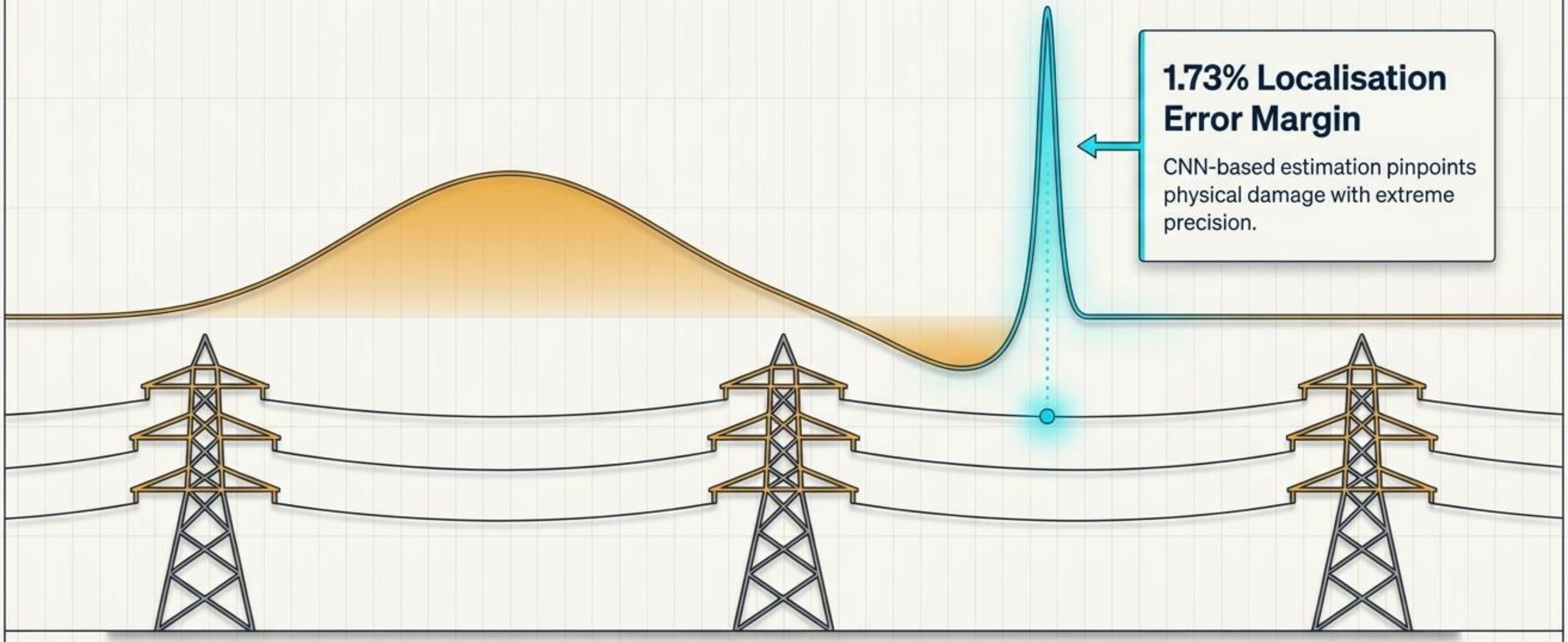
Reduced from 3130 violations to 11 in 10,000 cycles

Fault Classification Methodology



Attention weights filter out noise and prioritize specific time steps in the multi-modal data fusion to instantly identify the true fault signature.

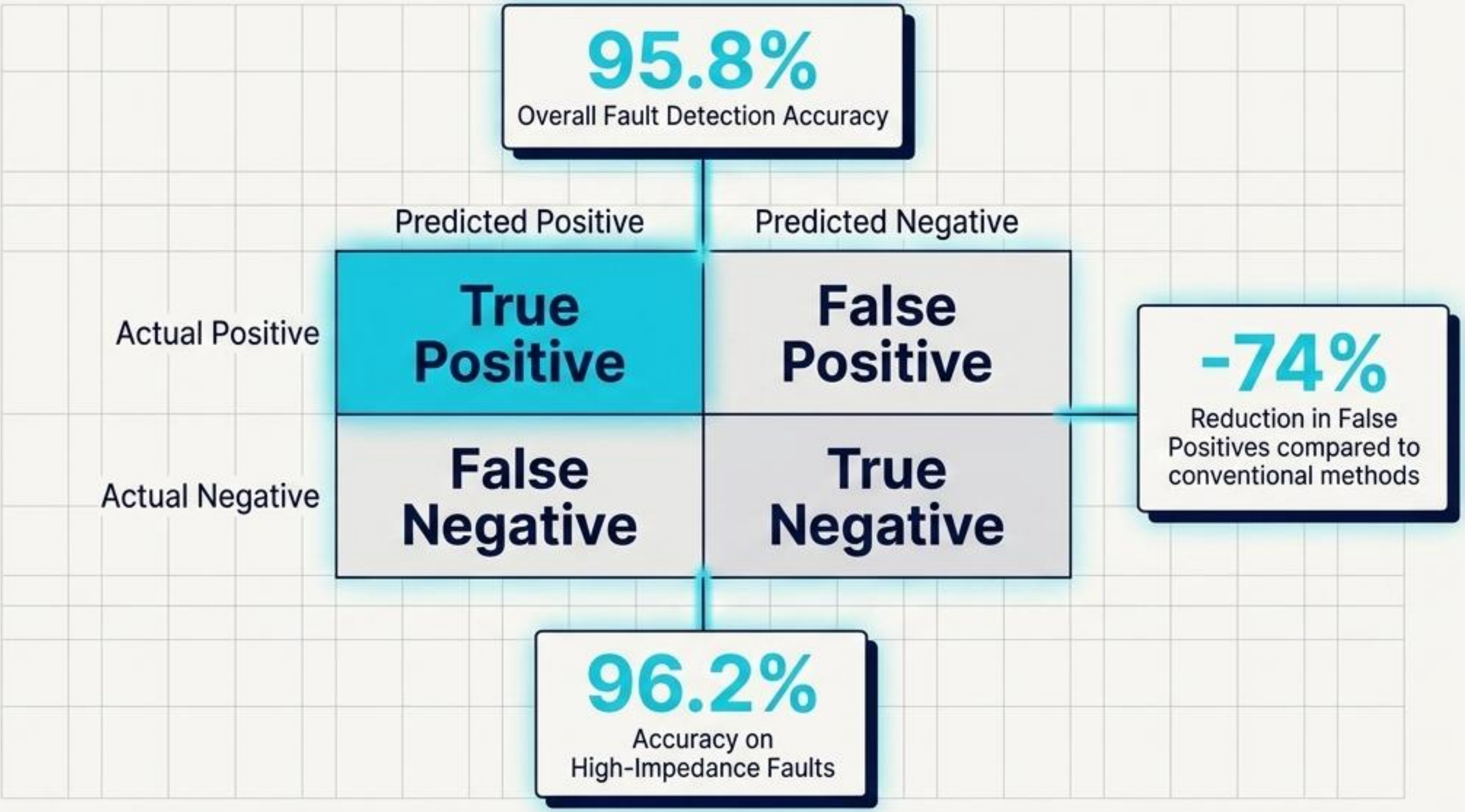
Fault Localization **(NN-Based Estimation)**



1.73% Localisation Error Margin

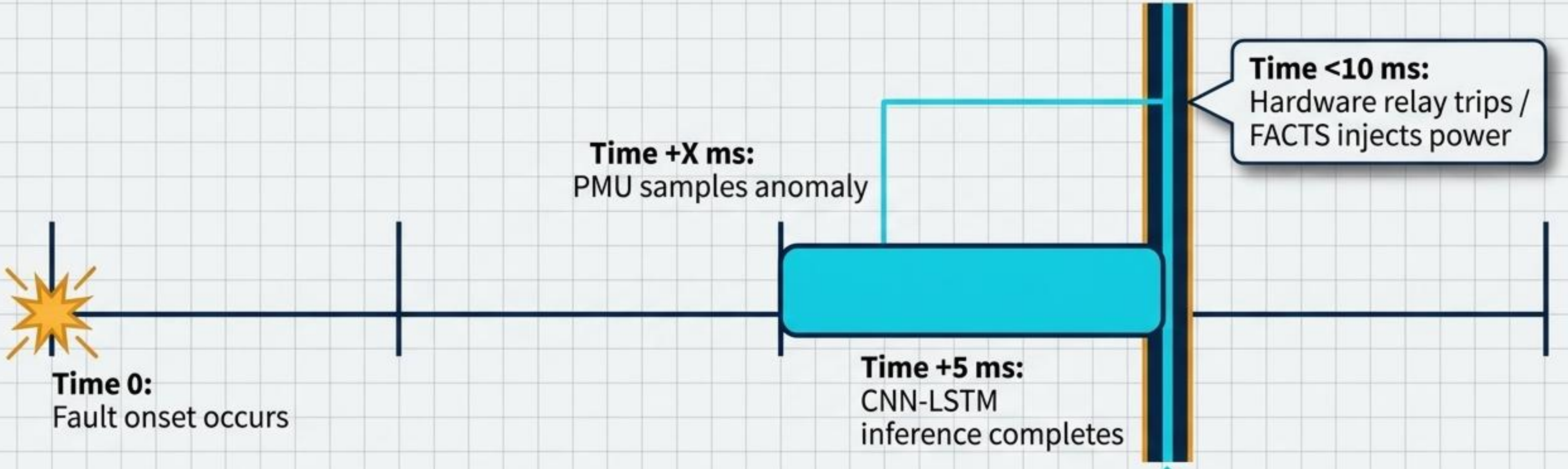
CNN-based estimation pinpoints physical damage with extreme precision.

Fault Detection Accuracy Results



Extensive offline performance data confirms the algorithmic accuracy, establishing a flawless baseline before transitioning the system to physical hardware stress-testing.

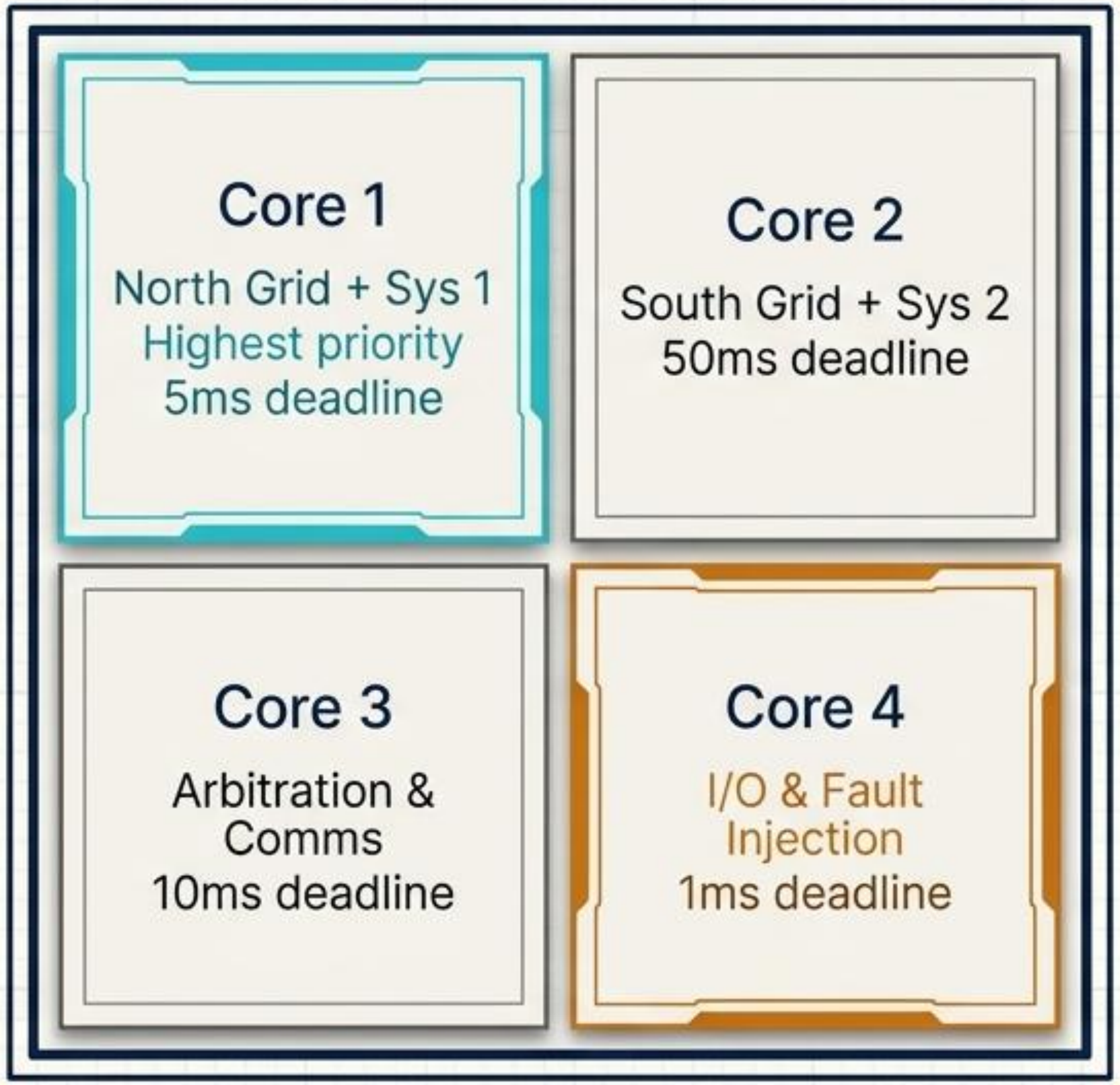
Real-Time Fault Response (<10ms on HIL)



61.4% faster response versus the best traditional rule-based methods.

HIL Fault Validation Setup

Deploying CAPSM on Real-Time Hardware: The OPAL-RT Controller-HIL Setup

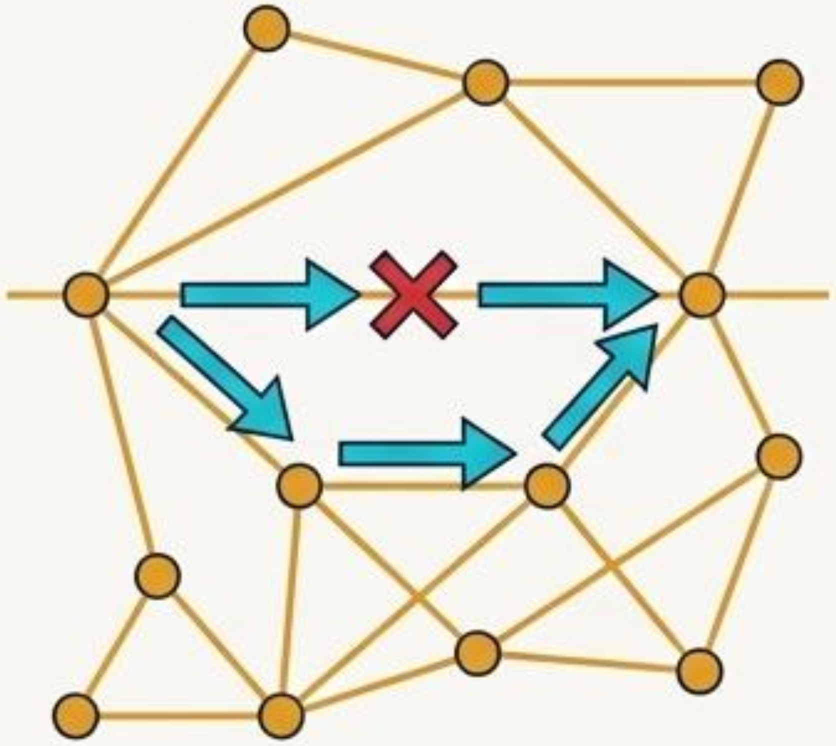


Solver Configuration

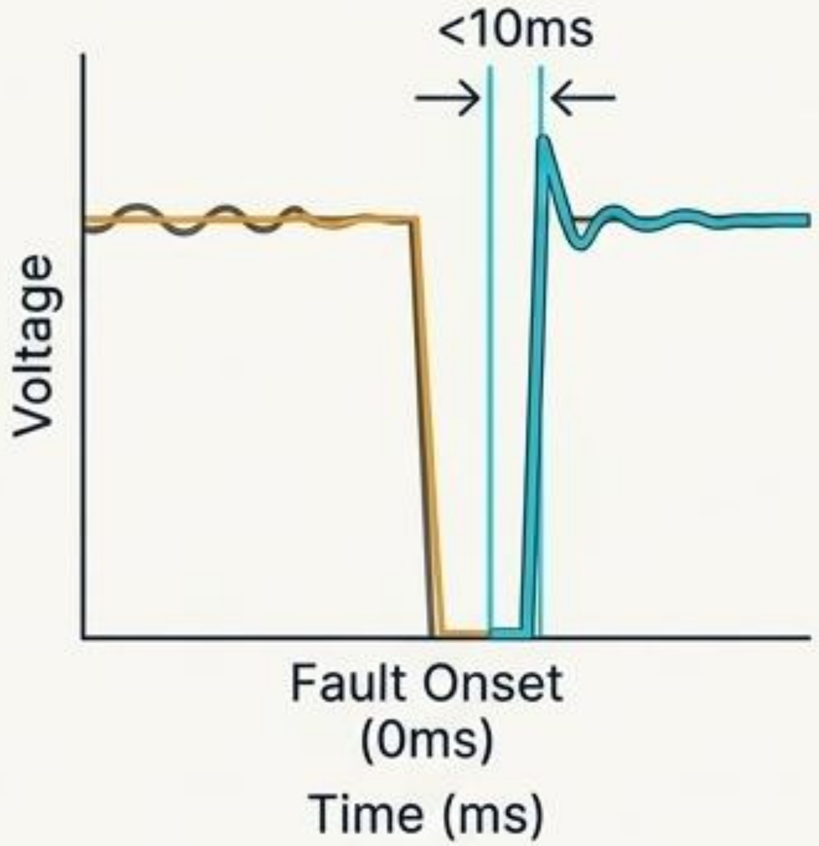
ARTEMIS solver running at a 50μs Electromagnetic Transient (EMT) time step.

Fault Scenario Testing

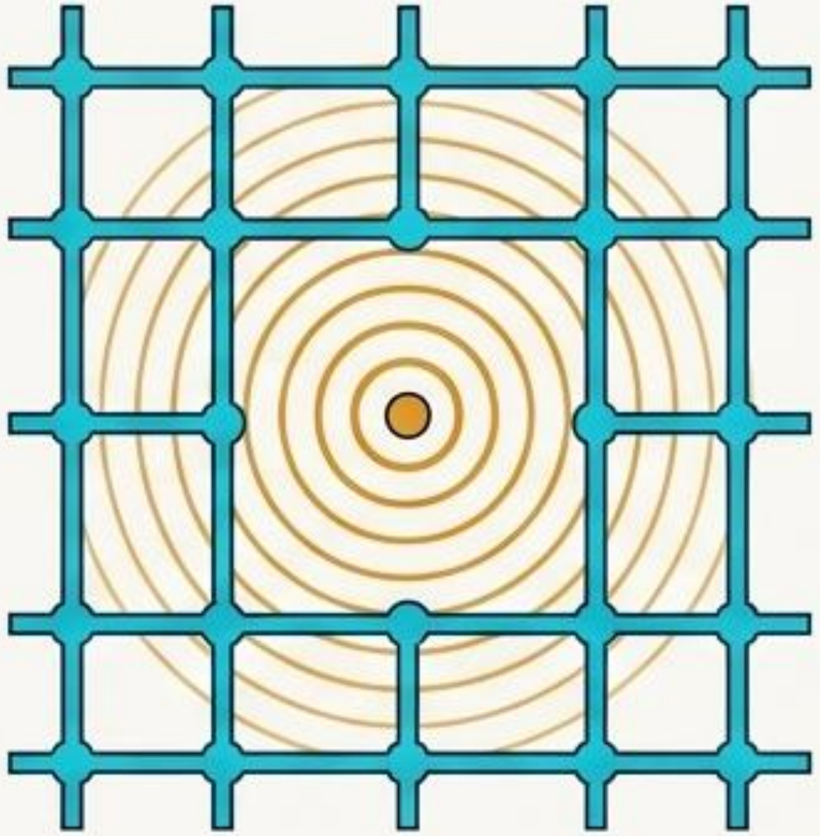
Scenario 1:
Line Tripping



Scenario 2:
Three-Phase Bus Fault



Scenario 3:
Cascading Failures



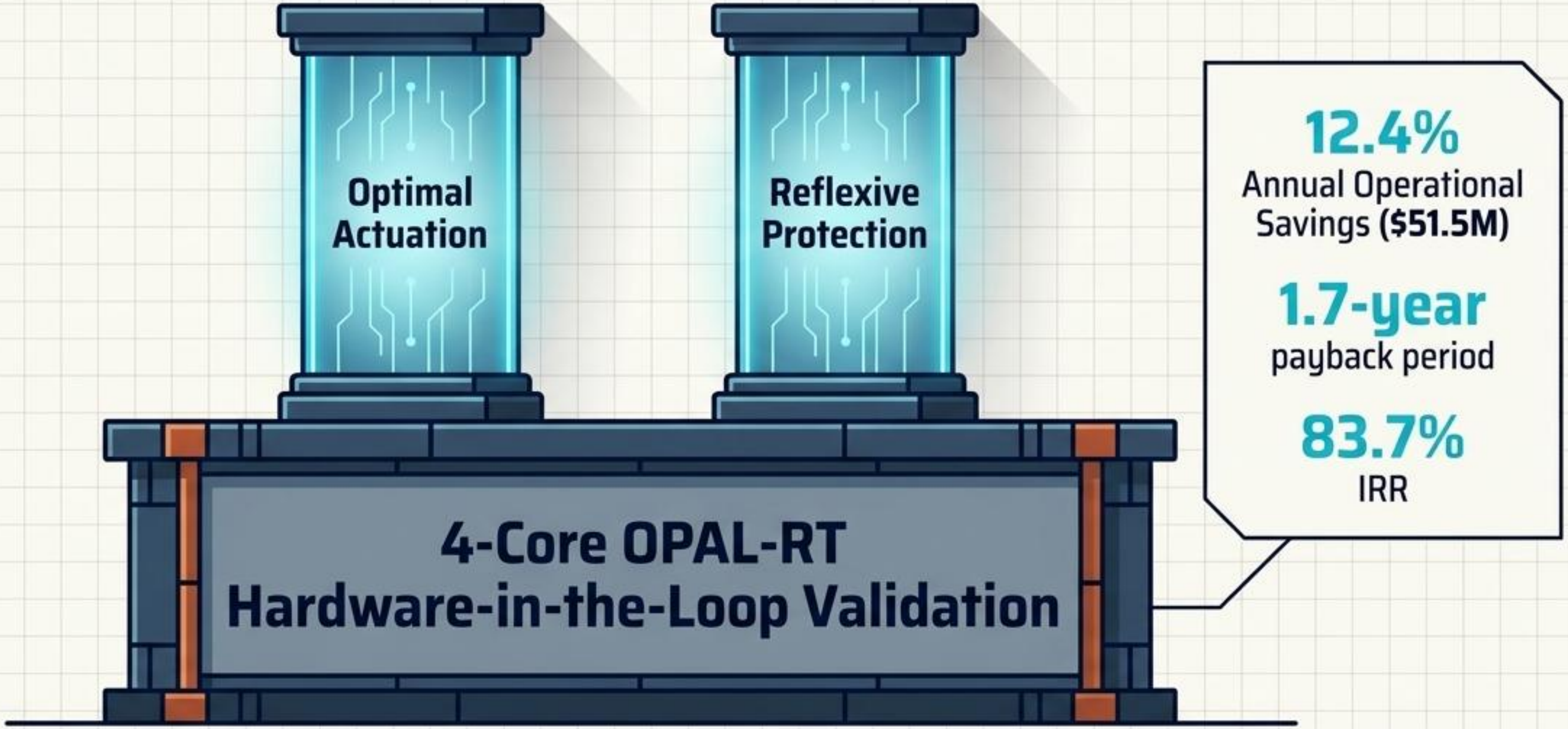
Hardware-in-the-Loop stress-testing proves that the system responds correctly and autonomously to extreme physical damage.

Performance Metrics Comparison

Engineering Dimension	CAPSM PINN	Conventional Impedance Relays
Latency	Sub-10ms	20-40ms
Localisation Error	1.73%	5-10%
False Positives	-74% Reduction	Baseline
High-Impedance Faults	96.2% Sensitivity	Low / Variable
Bidirectional Flow	Native Handling	Requires Complex Elements

AI-driven protection vastly outperforms conventional physics-only relays across all critical engineering dimensions in DER-rich environments.

Summary of Contributions



CAPSM successfully bridges theoretical AI to practical power engineering, delivering a field-ready blueprint for the modern grid.